

UNIT 4

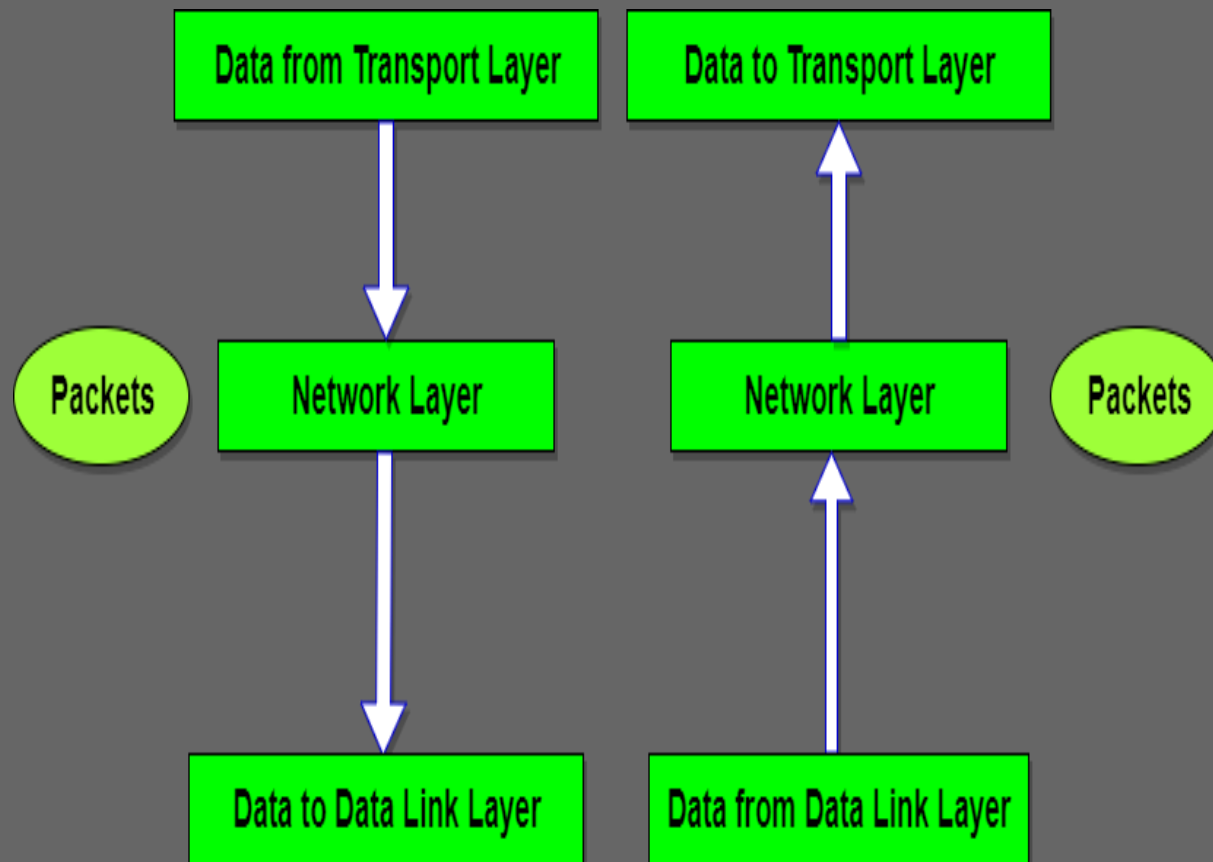
THE NETWORK LAYER

CONTENTS (8HRS)

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Introduction to Network Layer

- Layer-3 in the OSI model is called Network layer.
- Network layer manages options pertaining to host and network addressing, managing sub-networks, and internetworking.
- Network layer takes the responsibility for routing packets from source to destination within or outside a subnet.
- Network layer has the responsibility to route the packets from source to destination, mapping different addressing schemes and protocols.



4.1 Functions of Network Layer

- Addressing devices and networks.
- Populating routing tables or static routes.
- Queuing incoming and outgoing data and then forwarding them according to constraints set for those packets.
- Internetworking between two different subnets.
- Delivering packets to destination with best efforts.
- Provides connection oriented and connection less mechanism.

4.2 Virtual Circuits and Datagram Subnets

- Computer networks that provide connection-oriented service are called Virtual Circuits while those providing connection-less services are called as Datagram networks.

1. Virtual Circuits:

- It is connection-oriented, meaning that there is a reservation of resources like buffers, CPU, bandwidth, etc. for the time in which the newly setup VC is going to be used by a data transfer session.
- The first sent packet reserves resources at each server along the path. Subsequent packets will follow the same path as the first sent packet for the connection time.

- Since all packets follow a specific path, packets are received in-order at the destination.
- Virtual Circuit Switching ensures that all packets successfully reach the destination.
- From the above points, it can be concluded that Virtual Circuits are a highly reliable method of data transfer.
- The issue with virtual circuits is that each time a new connection is set up, resources and extra information have to be reserved at every router along the path, which becomes problematic if many clients are trying to reserve a routers resources simultaneously.

2. Datagram Networks:

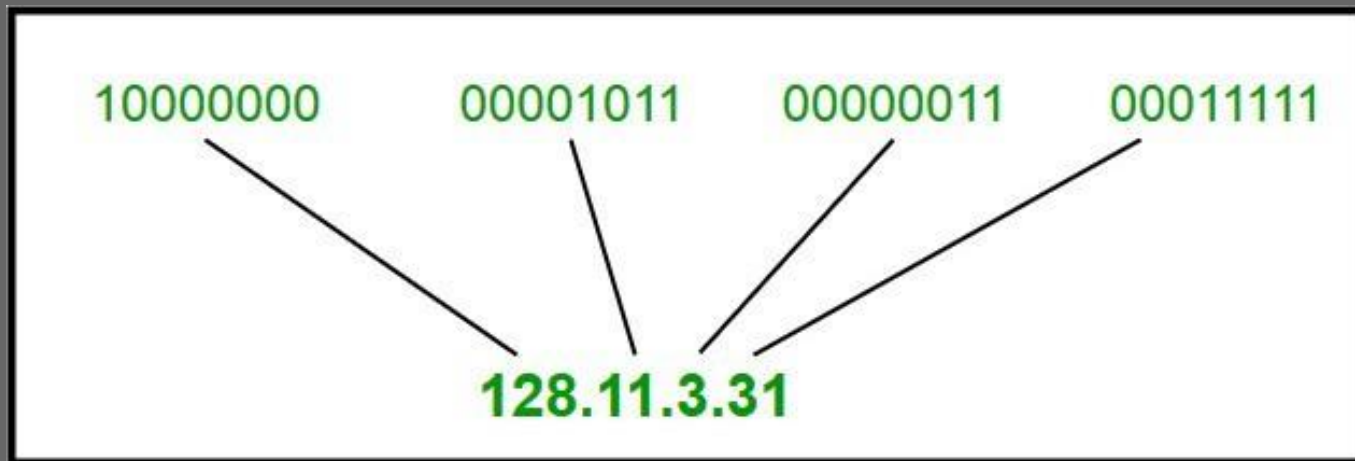
- It is a connection-less service. There is no need for reservation of resources as there is no dedicated path for a connection session.
- All packets are free to use any available path.
- Since every packet is free to choose any path, all packets must be associated with a header with proper information about the source and the upper layer data.
- The connection-less property makes data packets reach the destination in any order, which means that they can potentially be received out of order at the receiver's end.
- Datagram networks are not as reliable as Virtual Circuits.

- The major drawback of Datagram Packet switching is that a packet can only be forwarded if resources such as the buffer, CPU, and bandwidth are available. Otherwise, the packet will be discarded.
- But it is always easy and cost-efficient to implement datagram networks as there is no extra headache of reserving resources and making a dedicated each time an application has to communicate.

4.3 IPv4 Addresses:

- IP address is an address having information about how to reach a specific host, especially outside the LAN.
- An IP address is a 32 bit unique address having an address space of 2^{32} .

Dotted Decimal Notation:



Some points to be noted about dotted decimal notation:

- The value of any segment (byte) is between 0 and 255 (both included).
- There are no zeroes preceding the value in any segment (054 is wrong, 54 is correct).

Classful Addressing

The 32 bit IP address is divided into five sub-classes. These are:

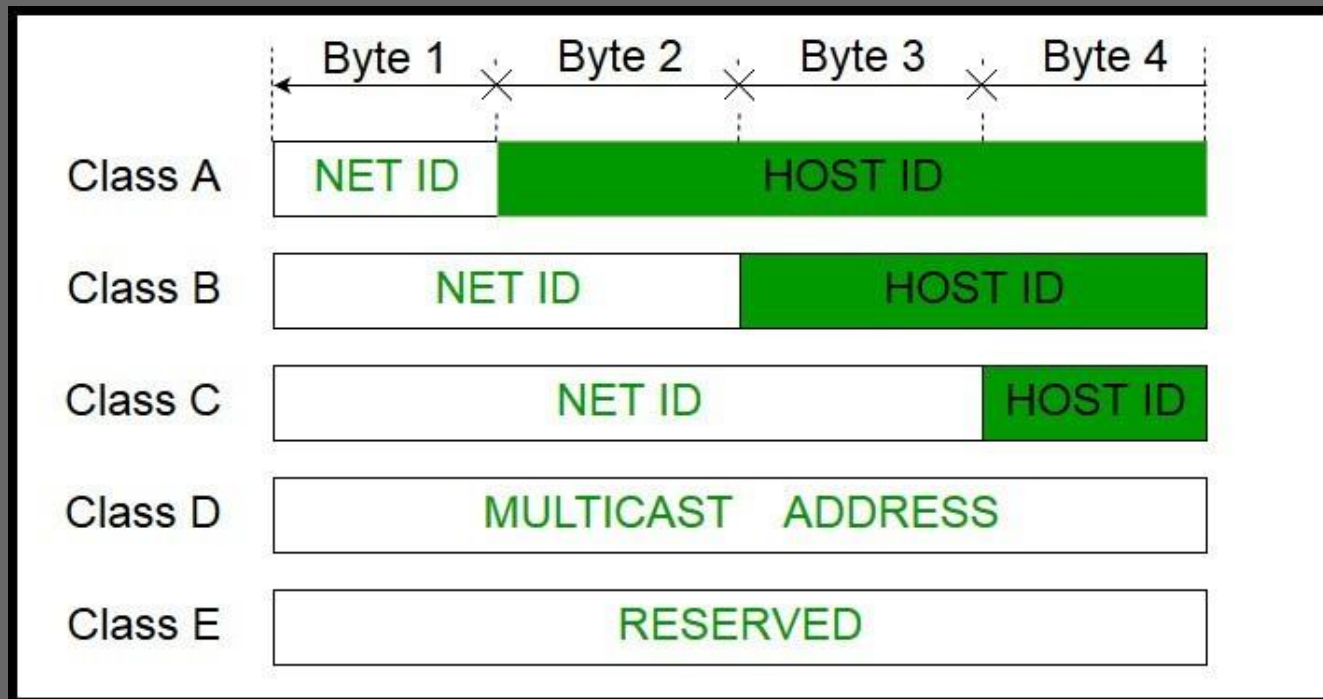
- Class A
- Class B
- Class C
- Class D
- Class E

Each of these classes has a valid range of IP addresses. Classes D and E are reserved for multicast and experimental purposes respectively. The order of bits in the first octet determine the classes of IP address.

IPv4 address is divided into two parts:

- Network ID
- Host ID

- The class of IP address is used to determine the bits used for network ID and host ID and the number of total networks and hosts possible in that particular class.
- Each ISP or network administrator assigns IP address to each device that is connected to its network.



- **Note:** IP addresses are globally managed by Internet Assigned Numbers Authority (IANA) and regional Internet registries (RIR).
- **Note:** While finding the total number of host IP addresses, 2 IP addresses are not counted and are therefore, decreased from the total count because the first IP address of any network is the network number and whereas the last IP address is reserved for broadcast IP.

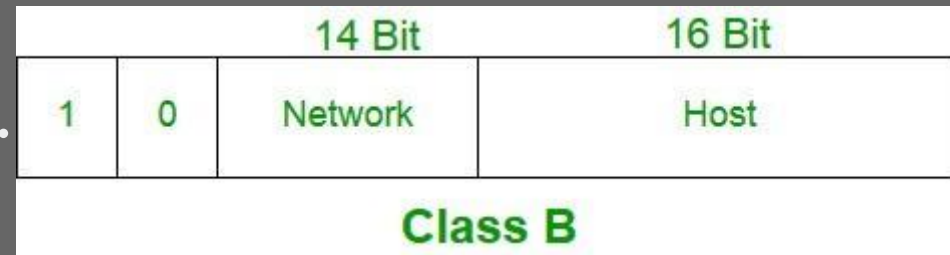
CLASS A ADDRESS:

- IP address belonging to class A are assigned to the networks that contain a large number of hosts.
- The network ID is 8 bits long.
- The host ID is 24 bits long.
- The higher order bit of the first octet in class A is always set to 0. The remaining 7 bits in first octet are used to determine network ID. The 24 bits of host ID are used to determine the host in any network. The default subnet mask for class A is 255.x.x.x. Therefore, class A has a total of:
 - $2^7 - 2 = 126$ network ID (Here 2 address is subtracted because 0.0.0.0 and 127.x.y.z are special address.)
 - $2^{24} - 2 = 16,777,214$ host ID
- IP addresses belonging to class A ranges from 1.x.x.x - 126.x.x.x

7 Bit		24 Bit
0	Network	Host
Class A		

CLASS B ADDRESS:

- IP address belonging to class B are assigned to the networks that ranges from medium-sized to large-sized networks.
- The network ID is 16 bits long.
- The host ID is 16 bits long.
- The higher order bits of the first octet of IP addresses of class B are always set to 10. The remaining 14 bits are used to determine network ID. The 16 bits of host ID is used to determine the host in any network. The default sub-net mask for class B is 255.255.x.x. Class B has a total of:
 - $2^{14} = 16384$ network address
 - $2^{16} - 2 = 65534$ host address
- IP addresses belonging to class B ranges from 128.0.x.x - 191.255.x.x.



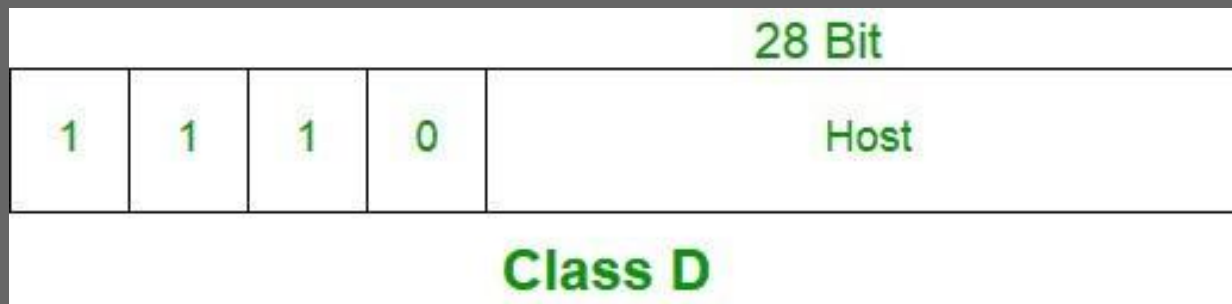
CLASS C ADDRESS:

- IP address belonging to class C are assigned to small-sized networks.
- The network ID is 24 bits long.
- The host ID is 8 bits long.
- The higher order bits of the first octet of IP addresses of class C are always set to 110. The remaining 21 bits are used to determine network ID. The 8 bits of host ID is used to determine the host in any network. The default sub-net mask for class C is 255.255.255.x. Class C has a total of:
 - $2^{21} = 2097152$ network address
 - $2^8 - 2 = 254$ host address
- IP addresses belonging to class C ranges from 192.0.0.x - 223.255.255.x.



CLASS D ADDRESS:

- IP address belonging to class D are reserved for multi-casting. The higher order bits of the first octet of IP addresses belonging to class D are always set to 1110. The remaining bits are for the address that interested hosts recognize.
- Class D does not possess any sub-net mask. IP addresses belonging to class D range from 224.0.0.0 - 239.255.255.255.



CLASS E ADDRESS:

- IP addresses belonging to class E are reserved for experimental and research purposes. IP addresses of class E ranges from 240.0.0.0 - 255.255.255.254. This class doesn't have any sub-net mask. The higher order bits of first octet of class E are always set to 1111.



Range of special IP addresses:

- 169.254.0.0 - 169.254.0.16 : Link local addresses
- 127.0.0.0 - 127.0.0.8 : Loop-back addresses
- 0.0.0.0 - 0.0.0.8 : used to communicate within the current network.

Rules for assigning Host ID:

Host ID's are used to identify a host within a network.

The host IDs are assigned based on the following rules:

- Within any network, the host ID must be unique to that network.
- Host ID in which all bits are set to 0 cannot be assigned because this host ID is used to represent the network ID of the IP address.
- Host ID in which all bits are set to 1 cannot be assigned because this host ID is reserved as a broadcast address to send packets to all the hosts present on that particular network.

Rules for assigning Network ID:

Hosts that are located on the same physical network are identified by the network ID, as all host on the same physical network is assigned the same network ID.

The network ID is assigned based on the following rules:

- The network ID cannot start with 127 because 127 belongs to class A address and is reserved for internal loop-back functions.
- All bits of network ID set to 1 are reserved for use as an IP broadcast address and therefore, cannot be used.
- All bits of network ID set to 0 are used to denote a specific host on the local network and are not routed and therefore, aren't used.

Summary of Classful addressing:

CLASS	LEADING BITS	NET ID BITS	HOST ID BITS	NO. OF NETWORKS	ADDRESSES PER NETWORK	START ADDRESS	END ADDRESS
CLASS A	0	8	24	2^7 (128)	2^{24} (16,777,216)	0.0.0.0	127.255.255.255
CLASS B	10	16	16	2^{14} (16,384)	2^{16} (65,536)	128.0.0.0	191.255.255.255
CLASS C	110	24	8	2^{21} (2,097,152)	2^8 (256)	192.0.0.0	223.255.255.255
CLASS D	1110	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	224.0.0.0	239.255.255.255
CLASS E	1111	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	240.0.0.0	255.255.255.255

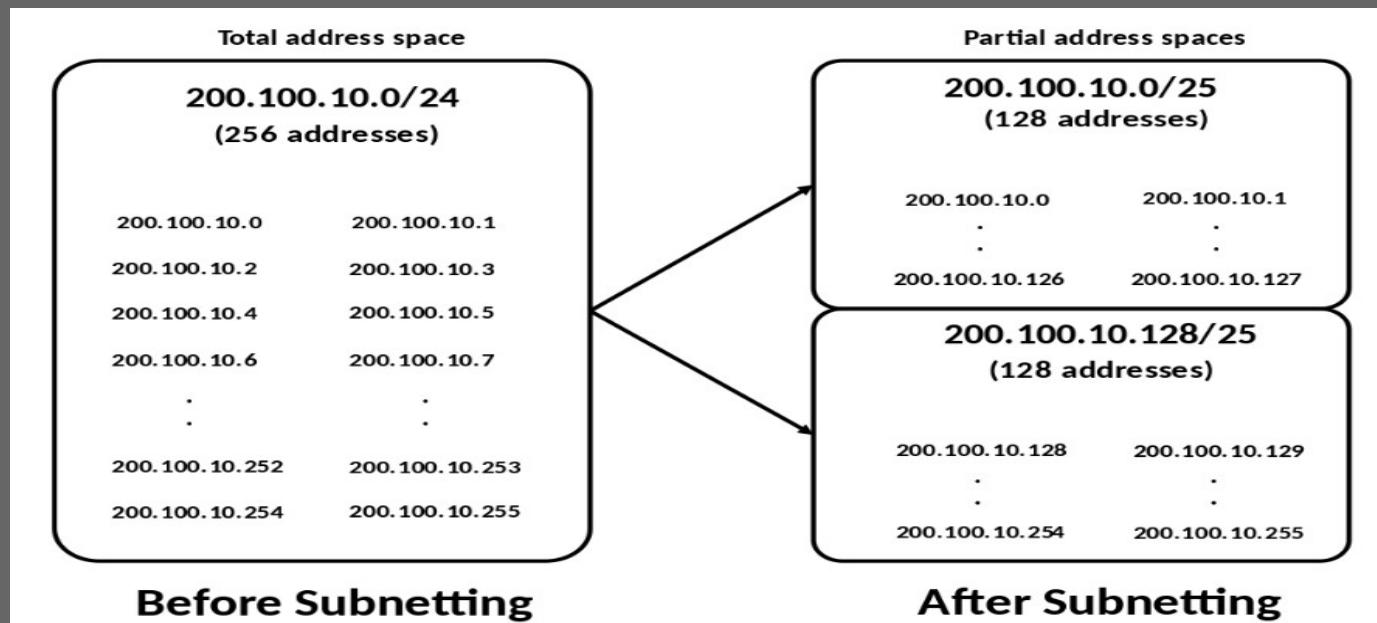
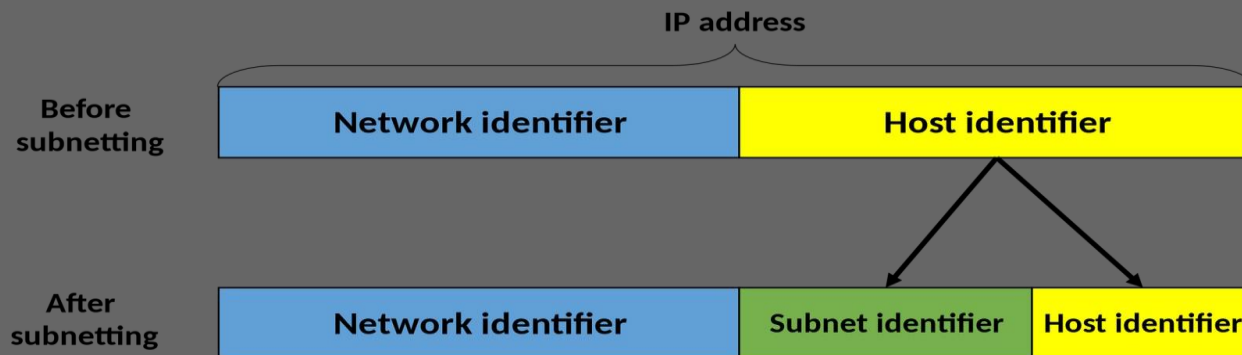
Problems with Classful Addressing:

- The problem with this classful addressing method is that millions of class A address are wasted, many of the class B address are wasted, whereas, number of addresses available in class C is so small that it cannot cater the needs of organizations.
- Class D addresses are used for multicast routing and are therefore available as a single block only. Class E addresses are reserved.

Since there are these problems, Classful networking was replaced by Classless Inter-Domain Routing (CIDR) in 1993.

Subnetting:

- A subnetwork or subnet is a logical subdivision of an IP network.
- The practice of dividing a network into two or more networks is called subnetting.



5 Subnetting Benefits:

1. Improve network performance and speed
2. Reduce network congestion
3. Boost network security
4. Control network growth
5. Ease of administration

What is Subnet Mask?

- A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s. In this way, the subnet mask separates the IP address into the network and host addresses.
- The “255” address is always assigned to a broadcast address, and the “0” address is always assigned to a network address. Neither can be assigned to hosts, as they are reserved for these special purposes.

SUBNET MASK IN IP ADDRESSING

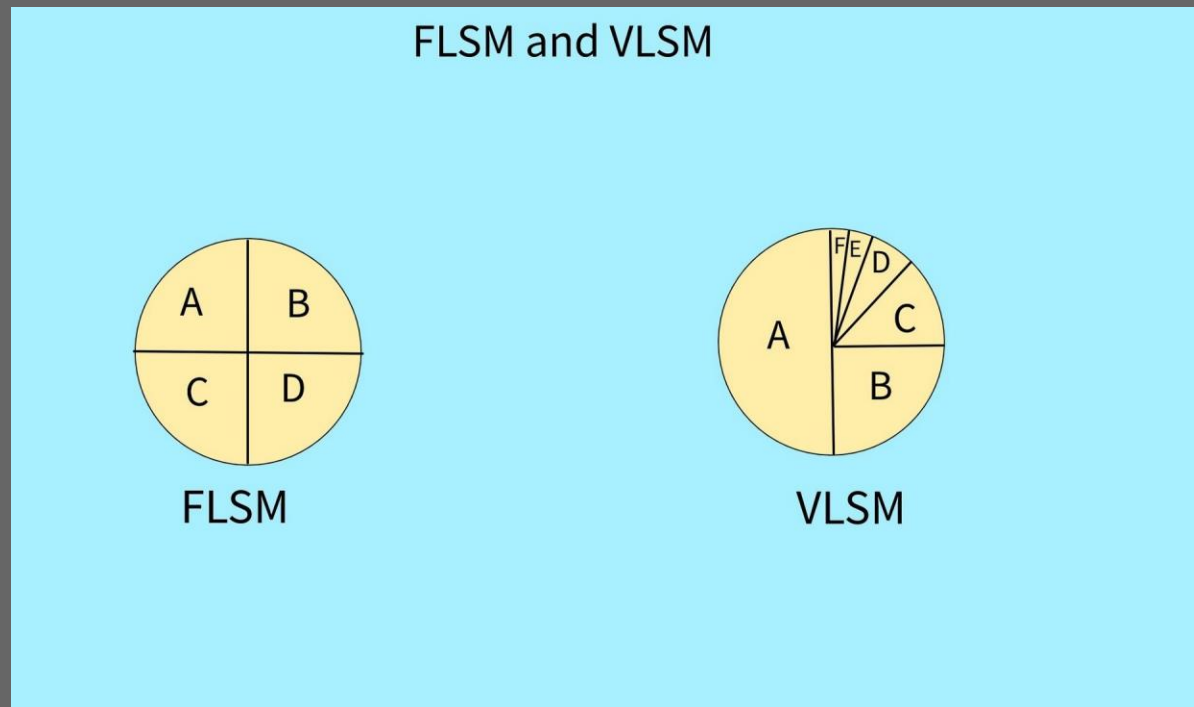
Class A	Network	Host	Host	Host
Subnet Mask	255	0	0	0

Class B	Network	Network	Host	Host
Subnet Mask	255	255	0	0

Class C	Network	Network	Network	Host
Subnet Mask	255	255	255	0

Type of Subnetting:

- There are two types of Subnetting FLSM(Fixed-Length Subnet Mask) and VLSM(Variable Length Subnet Mask). In FLSM, all subnets have equal number of host addresses and use same Subnet mask. In VLSM, subnets have flexible number of host addresses and use different subnet mask.
- Following figure shows an example of FLSM and VLSM.



Classless Inter-Domain Routing (CIDR):

- Classless inter-domain routing (CIDR) is a set of Internet protocol (IP) standards that is used to create unique identifiers for networks and individual devices.
- The IP addresses allow particular information packets to be sent to specific computers.
- Shortly after the introduction of CIDR, technicians found it difficult to track and label IP addresses, so a notation system was developed to make the process more efficient and standardized.
- That system is known as CIDR notation.

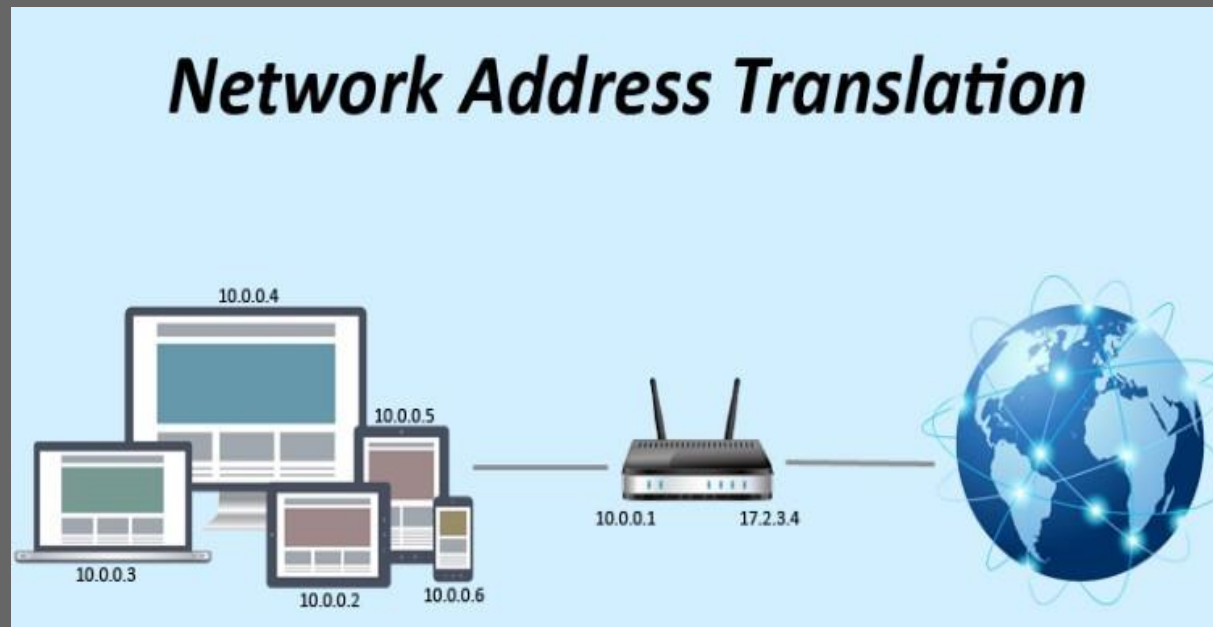
IP-Adresse	10 . 217 . 123 . 7 00001010 11011001 01111011 00000111
Subnetz- maske	255 . 255 . 240 . 0 11111111 11111111 11110000 00000000
Anzahl der Bits in der Subnetz- maske (Einsen)	$8 + 8 + 4 + 0 = 20$
IP-Adresse in CIDR-Notation	10.217.123.7/20

SUBNETTING NUMERICALS IN
WHITEBOARD

PRACTICE PRACTICE PRACTICE

Network Address Translation (NAT):

- To access the Internet, one public IP address is needed, but we can use a private IP address in our private network.
- The idea of NAT is to allow multiple devices to access the Internet through a single public address.



- To achieve this, the translation of private IP address to a public IP address is required.

- Network Address Translation (NAT) is a process in which one or more local IP address is translated into Global IP address and vice versa in order to provide Internet access to the local hosts.
- Also, it does the translation of port numbers i.e. masks the port number of the host with another port number, in the packet that will be routed to the destination.
- It then makes the corresponding entries of IP address and port number in the NAT table.
- NAT generally operates on router or firewall.

Advantages of NAT:

- Conserves legally registered addresses.
- Increases flexibility when connecting to the internet.

Disadvantages of NAT:

- Causes loss of end-to-end IP traceability.
- May cause delay in IPv4 communication.

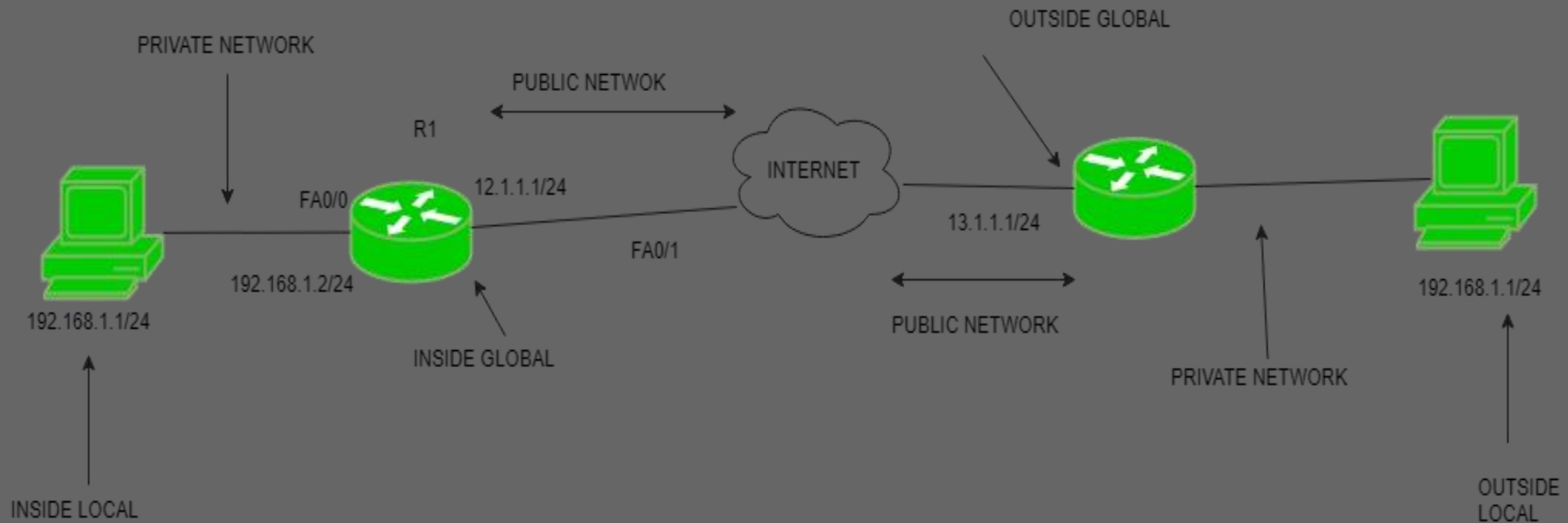
Network Address Translation (NAT) Types:

There are 3 types of NAT: Static NAT, Dynamic NAT and Overloading

- Static NAT (one-to-one): This type of NAT is designed to allow one-to-one mapping between local and global addresses. It requires one real Internet IP address for every host on your network.
- Dynamic NAT (many-to-many): This version gives the ability to map an unregistered IP address to a registered IP address from out of a pool of registered IP addresses.
- Overloading (one-to-many): This is the most popular type of NAT configuration. It is form of dynamic NAT that maps multiple unregistered IP addresses to a single registered IP address (many-to-one) by using different source ports. It's also known as Port Address Translation (PAT) which is commonly referred to as NAT Overload. NAT Overload is the real reason we haven't run out of valid IP addresses on the Internet.

NAT inside and outside addresses:

- Inside refers to the addresses which must be translated.
- Outside refers to the addresses which are not in control of an organization



- Inside local address: An IP address that is assigned to a host on the Inside (local) network. These are private IP address. This is the inside host seen from the inside network.
- Inside global address: IP address that represents one or more inside local IP addresses to the outside world. This is the inside host as seen from the outside network.
- Outside local address: This is the actual IP address of the destination host in the local network after translation.
- Outside global address: This is the outside host as seen from the outside network. It is the IP address of the outside destination host before translation.

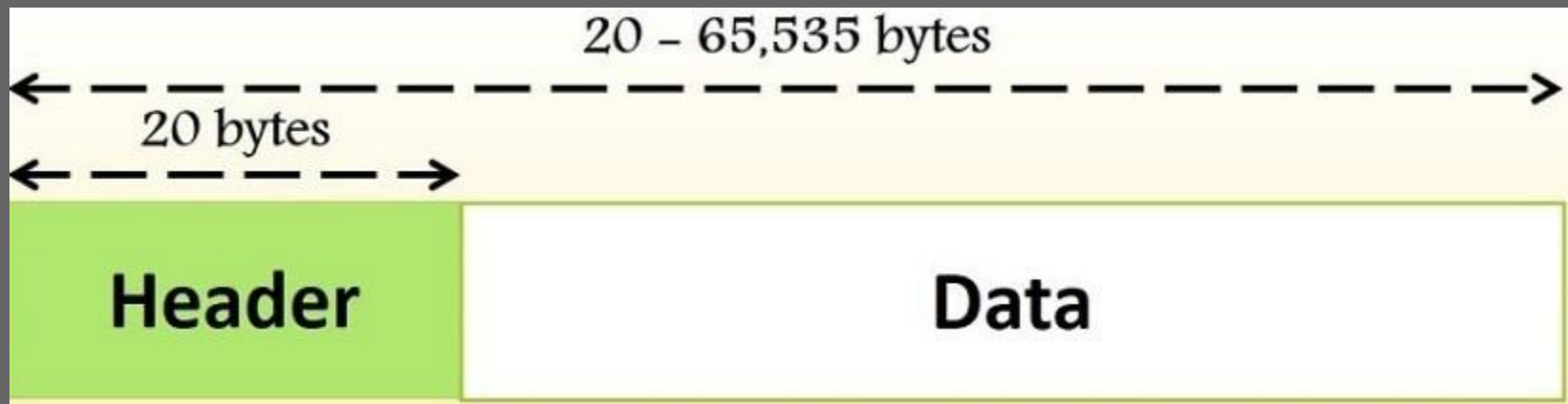
Network Address Translation (NAT) working:

- Generally, the border router is configured for NAT i.e the router which has one interface in local (inside) network and one interface in the global (outside) network. When a packet traverse outside the local (inside) network, then NAT converts that local (private) IP address to a global (public) IP address. When a packet enters the local network, the global (public) IP address is converted to a local (private) IP address.

4.4 IPv4 Datagram Format and Fragmentation

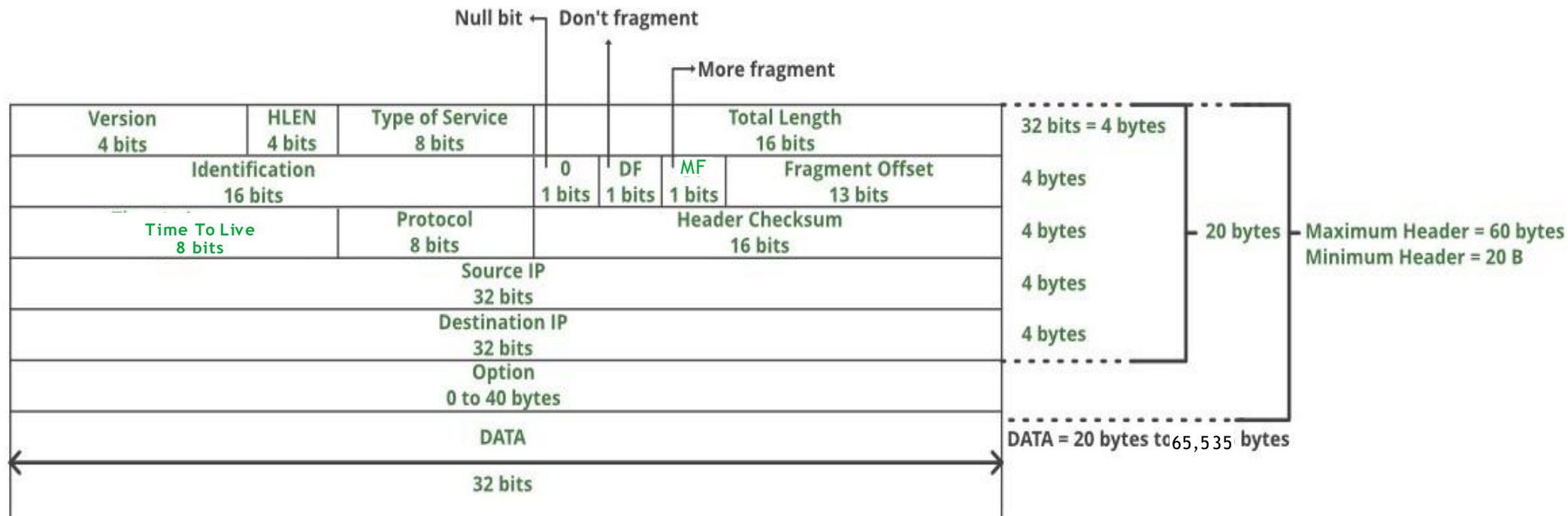
1. Datagram Formats:

- Packets in the network layer are called datagrams.
- A datagram is a variable-length packet consisting of two parts: header and data.
- The header is 20 to 60 bytes in length and contains information essential to routing and delivery.
- The following figure shows the IP datagram format.



IPv4 Datagram Header

- Size of the header is 20 to 60 bytes.



1. VERSION: Version of the IP protocol (4 bits), which is 4 for IPv4.
2. HLEN: IP header length is 4 bits. The minimum value for this field is 5 ($5 \times 4 = 20$) and the maximum is 15 ($15 \times 4 = 60$).

3. Type of service: Low Delay, High Throughput, Reliability (8 bits)
4. Total Length: Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.
5. Identification: Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits).
6. Flags: 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag and more fragments flag.
7. Fragment Offset: This offset tells the exact position (checkpoint) of the fragment in the original IP packet.
8. Time to live: Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.

9. Protocol: Name of the protocol to which the data is to be passed (8 bits)

10. Header Checksum: 16
bits header checksum for checking
errors in the datagram header

11. Source IP address: 32 bits IP address of the sender

12. Destination IP address: 32 bits IP address of the receiver

13. Option: Used by the Network administrator to check whether a path is working or not. These options may contain values for options such as traceroute etc.

14. Data: It contains the transport-layer segment to be delivered to the destination.

Due to the presence of options, the size of the datagram header can be of variable length (20 bytes to 60 bytes).

Datagram Fragmentation:

- Different Networks may have different maximum transmission unit (MTU), for example due to differences in LAN technology. When one network wants to transmit datagrams to a network with a smaller MTU, the routers on path may fragment and reassemble datagrams.
- When a packet is received at the router, destination address is examined and MTU is determined. If size of the packet is bigger than the MTU, and the 'Do not Fragment (DF)' bit is set to 0 in header, then the packet is fragmented into parts and sent one by one. The maximum size of each fragment is the MTU minus the IP header size (Minimum 20 bytes and Maximum 60 bytes).

Example: For a data packet of 4000 bytes and MTU of 1500 bytes, we have actual data of 3980 bytes that is to be transmitted and 1480 bytes is the maximum data size that is permissible to be sent. So, there would be 3 fragments:

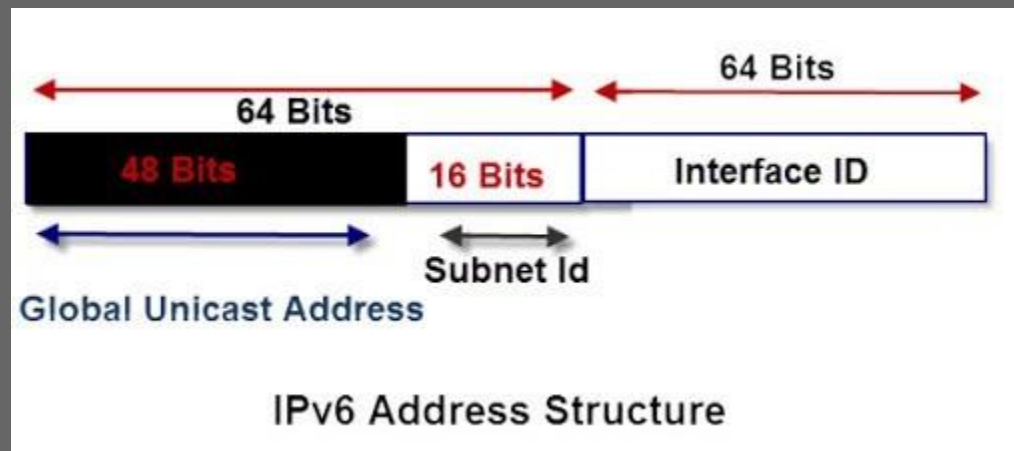
- For the first fragment, data size = 1480 bytes, offset = 0 and MF(more fragment) flag = 1
- For the second fragment, data size = 1480 bytes, offset = 1480 and MF flag = 1
- For the third fragment, data size = 1020 bytes, offset = 2960 and MF flag = 0

All fragments would be having same identification number, thus indicating that all the fragments belong to the same parent data packet.

4.5 IPv6 Address Structure and advantages over IPv4

IPv6 Addressing and its Features:

- IPv6 was developed by Internet Engineering Task Force (IETF) to deal with the problem of IP v4 exhaustion.
- IPv6 is 128-bits address having an address space of 2^{128} , which is way bigger than IPv4.
- In IPv6 we use Colon-Hexa representation.



Features of IPv6 Addressing:

1. Larger Address Space

In contrast to IPv4, IPv6 uses 4 times more bits to address a device on the Internet. This much of extra bits can provide approximately 3.4×10^{38} different combinations of addresses. This address can accumulate the aggressive requirement of address allotment for almost everything in this world. According to an estimate, 1564 addresses can be allocated to every square meter of this earth.

2. Simplified Header

IPv6's header has been simplified by moving all unnecessary information and options (which are present in IPv4 header) to the end of the IPv6 header. IPv6 header is only twice as bigger than IPv4 provided the fact that IPv6 address is four times longer.

3. End-to-end Connectivity

Every system now has unique IP address and can traverse through the Internet without using NAT or other translating components. After IPv6 is fully implemented, every host can directly reach other hosts on the Internet.

4. Faster Forwarding/Routing

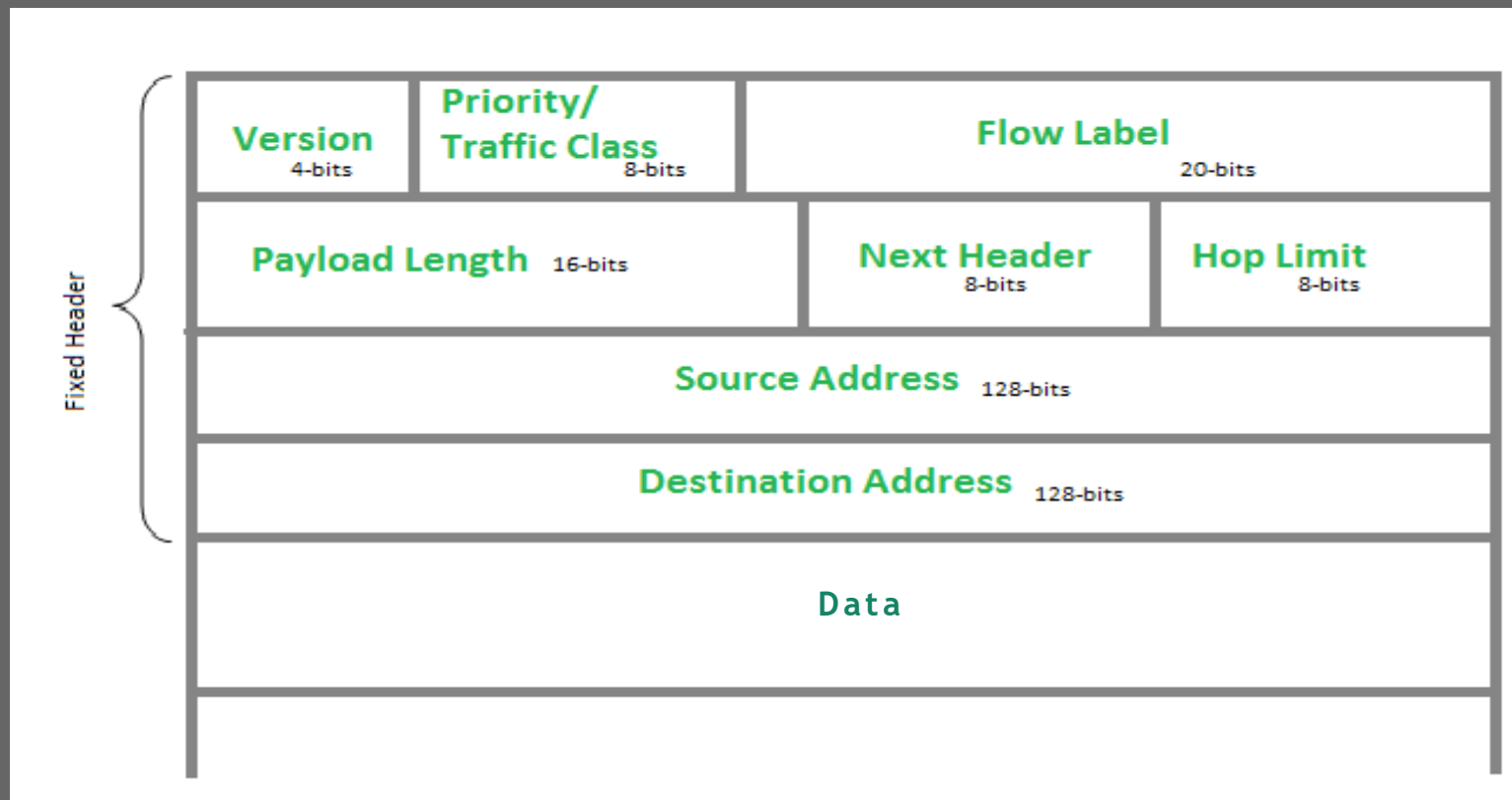
Simplified header puts all unnecessary information at the end of the header. The information contained in the first part of the header is adequate for a Router to take routing decisions, thus making routing decision as quickly as looking at the mandatory header.

5. IPSec

Initially it was decided that IPv6 must have IPSec security (provide data authentication, integrity, and confidentiality). This feature has now been made optional.

IPv6 Datagram Format

- IP version 6 is the new version of Internet Protocol, which is way better than IP version 4 in terms of complexity and efficiency.
- IP version 6 Header Format :



- Version (4-bits): Indicates version of Internet Protocol which contains bit sequence 0110.
- Priority/Traffic Class (8-bits): The Traffic Class field indicates class or priority of IPv6 packet which is similar to Service Field in IPv4 packet. It helps routers to handle the traffic based on priority of the packet. If congestion occurs on router then packets with least priority will be discarded.
- Flow Label (20-bits): This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a particular packet belongs to a specific flow of information. It is designed for streaming/real-time media.

- Payload Length (16-bits): This field is used to tell the routers how much information a particular packet contains in its payload. Payload is composed of Extension Headers and Upper Layer data. With 16 bits, up to 65535 bytes can be indicated; but if the Extension Headers contain Hop-by-Hop Extension Header, then the payload may exceed 65535 bytes.
- Next header (8-bits): This field identifies the protocol to which the contents of this datagram will be delivered (eg; to TCP or UDP). The field uses the same values as the protocol field in the IPv4 header.

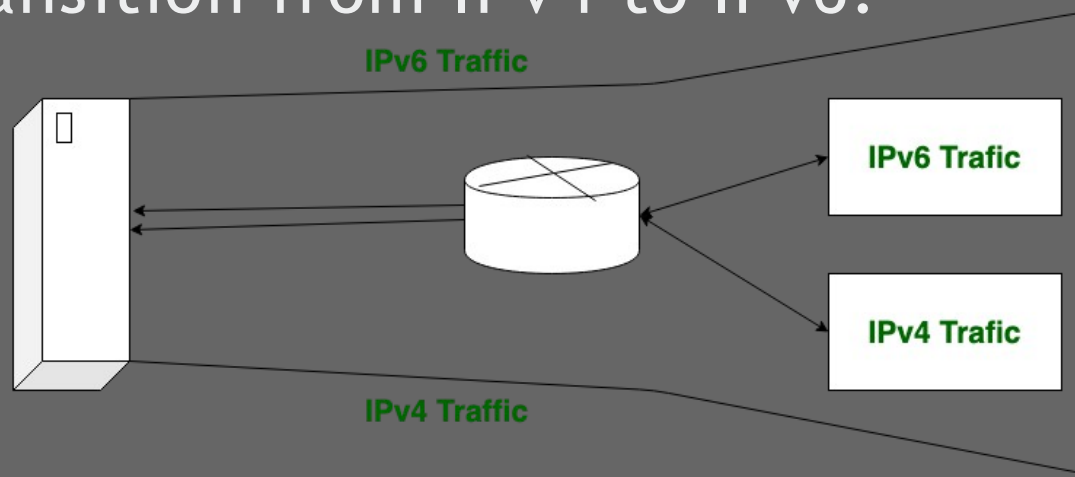
- Hop Limit (8-bits): This field is used to stop packet to loop in the network infinitely. This is same as TTL in IPv4. The value of Hop Limit field is decremented by 1 as it passes a link (router/hop). When the field reaches 0 the packet is discarded.
- Source Address (128-bits): This field indicates the address of originator of the packet.
- Destination Address (128-bits): This field provides the address of intended recipient of the packet.
- Data: This is the payload portion of the IPv6 datagram. When the datagram reaches its destination, the payload will be removed from the IP datagram and passed on to the protocol specified in the next header field.

Overview of IPv4 to IPv6 Transition Mechanisms:

When we want to send a request from an IPv4 address to an IPv6 address, but it isn't possible because IPv4 and IPv6 transition is not compatible. For a solution to this problem, we use some technologies. These technologies are Dual Stack Routers, Tunneling, and NAT Protocol Translation. These are explained as following below.

1. Dual-Stack Routers:

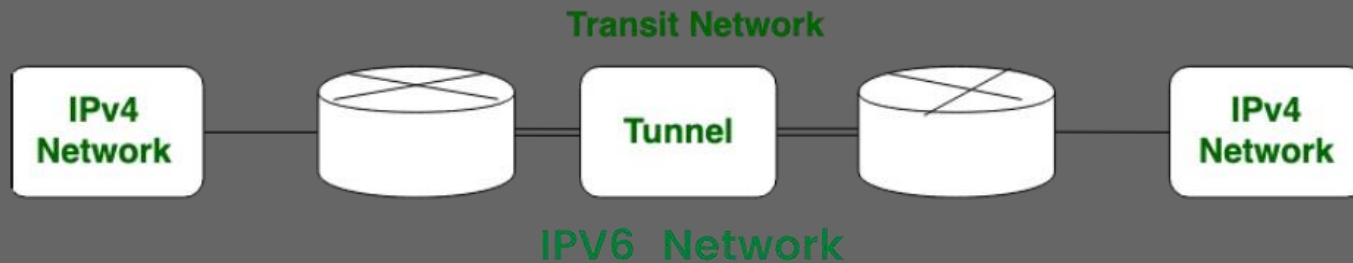
In dual-stack router, a router's interface is attached with IPv4 and IPv6 addresses configured are used in order to transition from IPv4 to IPv6.



In this above diagram, A given server with both IPv4 and IPv6 addresses configured can communicate with all hosts of IPv4 and IPv6 via dual-stack router (DSR). The dual stack router (DSR) gives the path for all the hosts to communicate with the server without changing their IP addresses.

2. Tunneling:

Tunneling is used as a medium to communicate the transit network with the different IP versions.

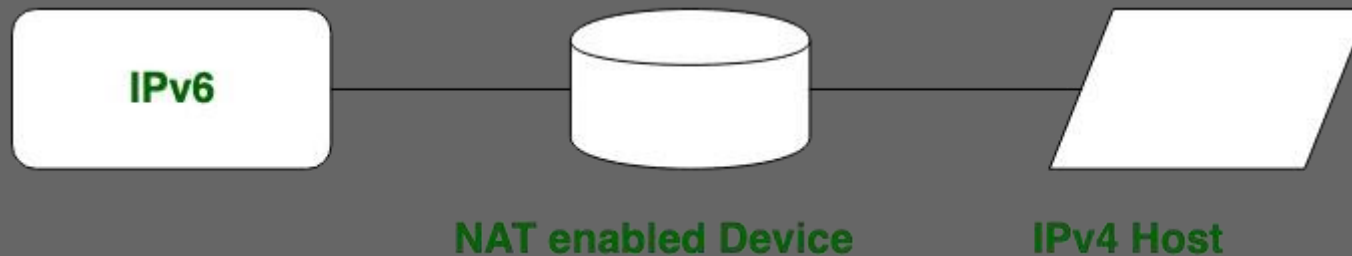


In this above diagram, the different IP versions such as IPv4 and IPv6 are present. The IPv4 networks can communicate with the transit or intermediate network on IPv6 with the help of the Tunnel. It's also possible that the IPv6 network can also communicate with IPv4 networks with the help of a Tunnel.

(IPv4 is encapsulated by 1st router in fig such that IPv6(IPv4) then decapsulated by 2nd router or simply remove IPv6 tag from IPv6(IPv4) to output IPv4 from 2nd router)

3. NAT Protocol Translation:

With the help of the NAT Protocol Translation technique, the IPv4 and IPv6 networks can also communicate with each other which do not understand the address of different IP version.

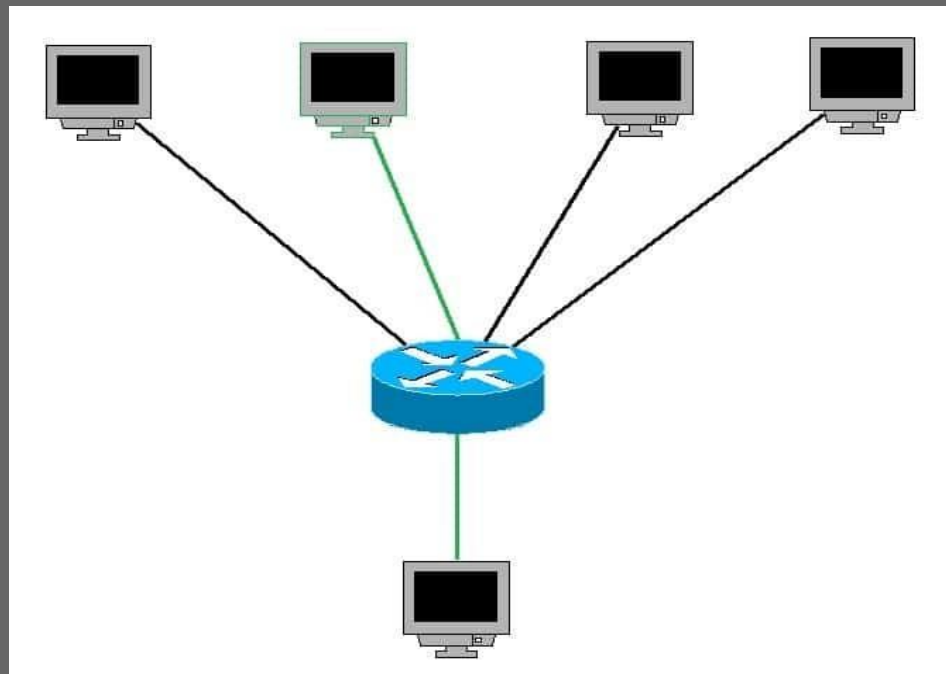


In the above diagram, an IPv4 address communicates with the IPv6 address via a NAT-PT device to communicate easily. In this situation, the IPv6 address understands that the request is sent by the same IP version (IPv6) and it responds.

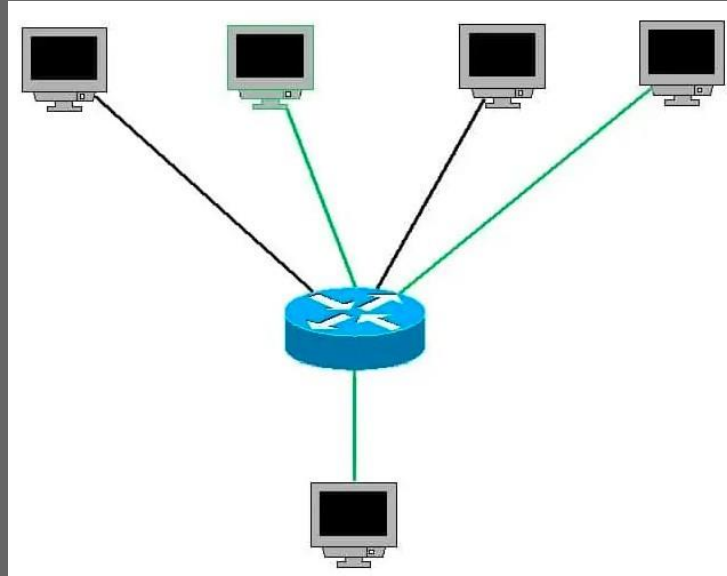
Example Addresses: Unicast, Multicast and Broadcast

IPv4 supports three different types of addressing modes namely unicast, multicast and broadcast.

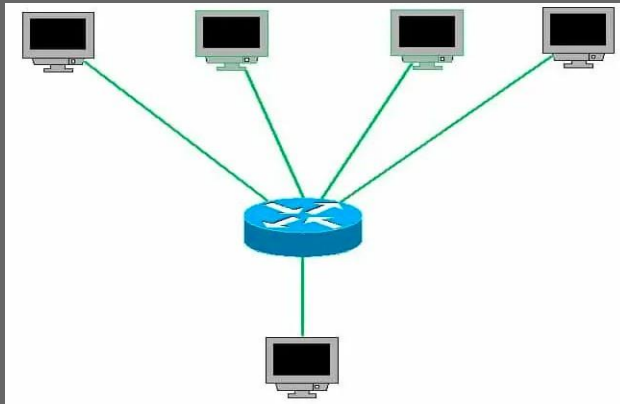
1. Unicast Addressing: Unicast addresses represent a single LAN interface. A unicast frame will be sent to a specific device, not to a group of devices on the LAN.



2. Multicast addresses: Multicast addresses represent a group of devices in a LAN. A frame sent to a multicast address will be forwarded to a group of devices on the LAN.



3. Broadcast addresses: Broadcast addresses represent all device on the LAN. Frames sent to a broadcast address will be delivered to all devices on the LAN.



4.6 Routing Algorithms: Distance Vector Routing, Link State Routing

- Routing is a process which is performed by layer 3 (or network layer) devices in order to deliver the packet by choosing an optimal path from one network to another.

There are 2 types of routing:

1. Static routing: Static routing is a process in which we have to manually add routes in routing table.

Advantage:

- No routing overhead for router CPU which means a cheaper router can be used to do routing.

Disadvantage:

- For a large network, it is a hectic task for administrator to manually add each route for the network in the routing table on each router.

2. Dynamic Routing:

- Dynamic routing makes automatic adjustment of the routes according to the current state of the route in the routing table. Dynamic routing uses protocols to discover network destinations and the routes to reach it. RIP and OSPF are the best examples of dynamic routing protocol.

Advantages:

- More effective at selecting the best route to a destination remote network.

Disadvantage:

- Consumes more bandwidth for communicating with other neighbors.

Interior vs Exterior Dynamic Routing:

- Interior routing protocols are designed for use within a contained network of limited size, whereas exterior routing protocols are designed to link multiple networks together.
- They can be used in combination in order to simplify network administration. For example, a network can be built with only border routers of a network running the exterior routing protocol, while all the routers on the network run the interior protocol, which prevents them from connecting outside the network without passing through the border.
- Nearly all routing protocols are interior routing protocols. Only BGP is commonly used as an exterior routing protocol.

Distance Vector vs Link State Dynamic Routing Protocol:

1. Distance Vector Routing Protocol:

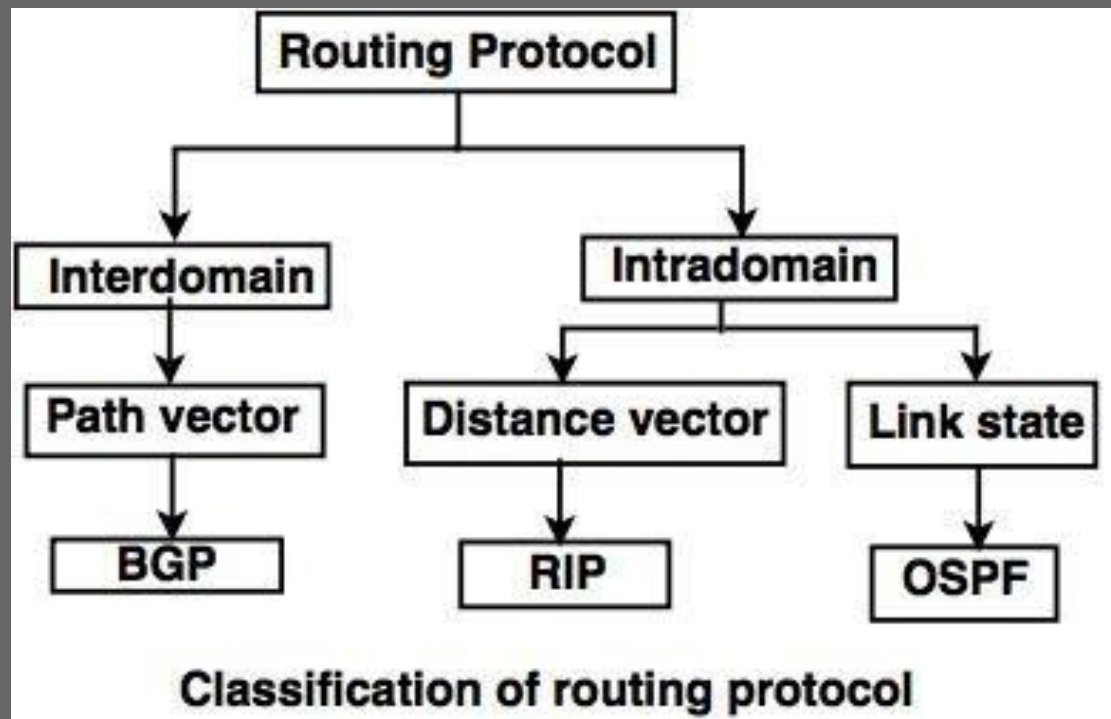
- It is dynamic routing algorithm in which each router computes distance between itself and each possible destination i.e; its immediate neighbors.
- The router share its knowledge about the whole network to its neighbors and accordingly updates table based on its neighbors.
- It makes use of Bellman Ford Algorithm for making routing tables.
- Distance vector protocols send their entire routing table to directly connected neighbors.
- Examples of DVR protocols include RIP- Routing Information Protocol and IGRP- Interior Gateway Routing Protocol.

2. Link State Routing Protocols:

- It is a dynamic routing algorithm in which each router shares knowledge of its neighbors with every other router in the network.
- A router sends its information about its neighbors only to all the routers through flooding.
- It makes use of Dijkstra's Algorithm for making routing tables.
- Link state protocols are also called shortest-path-first protocols.
- Examples of Link state routing protocols include OSPF- Open Shortest Path First and IS-IS Intermediate System to Intermediate System.

4.7 Routing Protocols:

- In computer networks, the routing protocol specifies how routers communicate to select the routes for information or data transfer.
- Different types of routing protocols are as follows:



Open Shortest Path First (OSPF):

- Open Shortest Path First (OSPF) is a link-state routing protocol that is used to find the best path between the source and the destination router using its own Shortest Path First.
- OSPF is developed by Internet Engineering Task Force (IETF) as one of the Interior Gateway Protocol (IGP), i.e, the protocol which aims at moving the packet within a large autonomous system or routing domain.
- It is a network layer protocol which works on the port number 89.
- OSPF uses multicast address 224.0.0.5 for normal communication and 224.0.0.6 for update to designated router(DR)/Backup Designated Router (BDR).

Border Gateway Protocol (BGP):

- BGP stands for Border Gateway Protocol.
- It is a protocol used to advertise the set of networks that can be reached within an autonomous system. BGP enables this information to be shared with the autonomous system.
- BGP (Border Gateway Protocol) is a protocol that manages how packets are routed across the internet through the exchange of routing and reachability information between edge routers. BGP directs packets between autonomous systems - networks managed by a single enterprise or service provider.
- It is a transport layer protocol which works on the port number 179.

4.8 Internet Control Protocols: ARP, RARP, ICMP

1. Address Resolution Protocol (ARP):

- Address Resolution Protocol (ARP) is a procedure for mapping a dynamic Internet Protocol address (IP address) to a permanent physical machine address in a local area network (LAN). The physical machine address is also known as a Media Access Control or MAC address.
- The job of the ARP is essentially to translate 32-bit addresses to 48-bit addresses and vice-versa. This is necessary because in IP Version 4 (IPv4), the most common level of Internet Protocol (IP) in use today, an IP address is 32-bits long, but MAC addresses are 48-bits long.
- ARP works between network layers 2 and 3 of the Open Systems Interconnection model (OSI model). The MAC address exists on layer 2 of the OSI model, the data link layer, while the IP address exists on layer 3, the network layer.

2. Reverse Address Resolution Protocol (RARP):

- RARP (Reverse Address Resolution Protocol) is a protocol by which a physical machine in a local area network can request to learn its IP address from a gateway server's Address Resolution Protocol (ARP) table or cache.
- A network administrator creates a table in a local area network's gateway router that maps the physical machine (or Media Access Control - MAC address) addresses to corresponding Internet Protocol addresses.
- When a new machine is set up, its RARP client program requests from the RARP server on the router to be sent its IP address. Assuming that an entry has been set up in the router table, the RARP server will return the IP address to the machine which can store it for future use.

3. Internet Control Message Protocol (ICMP):

- ICMP is a network layer protocol used by routers, intermediary devices and hosts to communicate error information (invalid IP) or updates to other routers, intermediary devices and hosts.
- ICMP messages are sent in several scenarios. For example, if one device sends a message that is too large for the recipient to process, the recipient will drop that message and send an ICMP message back to the source. Another example is when the network gateway finds a shorter route for the message to travel on. When this happens, an ICMP message is sent, and the packet is redirected to the shorter route.
- ICMP is also used for network diagnostics, specifically the ping and traceroute terminal utilities.

Questions?